

MoCafe v2.00 — Physics and Algorithms

Abstract

This note collects the physics and the numerical algorithms behind MoCafe, the Monte-Carlo dust radiative-transfer code: photon transport and the peeling-off estimator, panchromatic (SED) transport by grey rescaling, the Lucy (1999) mean-intensity estimator, the two dust-emission solvers (Lucy+SEDust and Bjorkman & Wood), the Modified Random Walk, the galaxy source models, and the random-sampling routines (alias method, exponential disk, deprojected Sersic). It is a companion to the user guide, which covers installation, the input reference, and the outputs.

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1. Monte-Carlo radiative transfer

MoCafe propagates photon *packets* (weighted bundles of photons) through a dusty medium. A packet carries a position, a propagation direction, a weight w , a wavelength bin, and—in SED/emission mode—a physical luminosity L_p [erg s⁻¹]. The medium is described by an extinction coefficient $\kappa_{\text{ext}} = \kappa_{\text{sca}} + \kappa_{\text{abs}}$ (per unit length), a single-scattering albedo $a = \kappa_{\text{sca}} / \kappa_{\text{ext}}$, and a scattering phase function.

1.1 Free path and optical depth

The probability that a packet travels an optical depth τ before an interaction is $e^{-\tau}$, so a free path is drawn as

$$\tau = -\ln(1 - \xi), \quad \xi \in [0, 1), \quad (1)$$

and converted to a physical length by integrating κ_{ext} along the ray (Amanatides–Woo voxel traversal on the Cartesian grid, a ray–sphere plus DDA walk in the clumpy medium, and an octree neighbour walk on the AMR grid). The three ray tracers share one interface so scattering and peeling-off are grid-agnostic.

1.2 Forced first scattering and reduced weight

For optically thin media most packets would escape without interacting, wasting samples. With `par%use_reduced_wgt` the first interaction is *forced* inside the grid. If τ_0 is the total optical depth from the birth point to the far edge, the first free path is drawn from the *truncated* distribution

$$\tau = -\ln[1 - \xi(1 - e^{-\tau_0})], \quad (2)$$

and the packet weight is reduced by the probability that it would have interacted at all,

$$w \leftarrow w(1 - e^{-\tau_0}). \quad (3)$$

The escaping fraction $e^{-\tau_0}$ is accounted for by the direct (unscattered) peel-off, so no energy is lost. The $n_{\text{scatt}} = 0$ step is special-cased in the transport loop for this reason.

1.3 Peeling-off estimator

Rather than waiting for packets to reach an observer by chance, at every emission and scattering site MoCafe adds the *expected* contribution toward each observer (the “peeling-off” or next-event estimator). For a scattering at position $\mathbf{r}$ with incoming direction $\hat{\mathbf{k}}$, the contribution to observer o in direction $\hat{\mathbf{n}}_o$ at distance d_o is

$$\Delta I_o = \frac{wa\Phi(\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_o)}{4\pi} e^{-\tau_o} \frac{1}{d_o^2}, \quad (4)$$

where Φ is the (normalized) phase function and τ_o is the optical depth from $\mathbf{r}$ to the observer along $\hat{\mathbf{n}}_o$. For an internal all-sky observer (§9.) the same quantity is binned into the HEALPix pixel of the arrival direction instead of a tangent-plane image. The peel is computed through the `raytrace_to_edge` procedure pointer so that the Cartesian, clumpy, and AMR media reuse identical scattering/peeling code.

2. Scattering

The default phase function is Henyey–Greenstein with asymmetry g ,

$$\Phi_{\text{HG}}(\mu) = \frac{1-g^2}{2(1+g^2-2g\mu)^{3/2}}, \quad \mu = \cos \theta, \quad (5)$$

sampled analytically for the scattering angle. When a Mueller matrix is supplied (`par%scatt_mat_file`) the code carries the Stokes vector (I, Q, U, V) , samples $\cos \theta$ from the tabulated S_{11} with an alias table (§8.), and rotates/updates the Stokes vector through the scattering matrix. In that case the peel-off of Eq. (4) uses the matrix-weighted phase function.

3. Panchromatic (SED) transport

3.1 Grey rescaling

In SED mode a packet is assigned a wavelength bin λ at birth and keeps it until absorption. Instead of storing a separate opacity grid per wavelength, MoCafe stores the density (hence the optical-depth structure) at a single reference wavelength λ_{ref} and rescales per packet (Jonsson 2006-style),

$$s_{\text{ext}}(\lambda) = \frac{\kappa_{\text{ext}}(\lambda)}{\kappa_{\text{ext}}(\lambda_{\text{ref}})}, \quad \tau(\lambda) = s_{\text{ext}}(\lambda) \tau_{\text{ref}}. \quad (6)$$

The ray tracers integrate τ_{ref} and multiply by the packet's s_{ext} , so all three tracers are shared with the monochromatic path and memory does not grow with the number of wavelengths. The wavelength-dependent albedo $a(\lambda)$ and asymmetry $g(\lambda)$ come from the same `kext_file` table $(\lambda, a, \langle \cos \rangle, C_{\text{ext}}/H)$.

3.2 Wavelength sampling

The source spectrum L_λ (a Planck function at `par%tstar`, a tabulated `par%source_spectrum`, or per component in the multi-source case) is turned into a luminosity fraction per bin and sampled with an alias table. Each packet then carries a physical luminosity $L_p = L_{\text{tot}}/N_{\text{phot}}$, which makes the emission energy budget absolute (erg s^{-1}).

4. Mean intensity in each cell (Lucy 1999)

The heating of the dust is set by the mean intensity J_λ in each cell. MoCafe uses the Lucy (1999) *pathlength* estimator, which accumulates a contribution from every path segment a packet takes through a cell, not just from discrete interaction points:

$$J_\lambda = \frac{1}{4\pi V \Delta\lambda d_{\text{cm}}^2} \sum_{\text{segments in cell}} L_p w_\ell, \quad (7)$$

where ℓ is the length of the segment, V the cell volume, $\Delta\lambda$ the bin width, and d_{cm} the length unit in cm. Because every segment contributes, the estimator has far lower variance than a collision estimator, especially in optically thin cells. The absorbed power in a cell follows as

$$P_{\text{abs}} = 4\pi V \int \kappa_{\text{abs}}(\lambda) J_\lambda d\lambda, \quad (8)$$

and is written to the `_jlam` output together with a consistency check (the pathlength tally EABS_A versus an event counter EABS_B). The forced-first-flight segment is added analytically.

5. Dust thermal emission

Two solvers turn the absorbed energy into an emission spectrum. Both conserve energy exactly: the total emitted luminosity equals the total absorbed luminosity.

5.1 Mode 1: Lucy (1999) with SEDust

The radiation field J_λ from Eq. (7) is handed cell-by-cell to the SEDust grain library, which returns the emission spectrum ε_λ for the local field. SEDust integrates over the grain-size distribution and, for small grains and PAHs, over the *temperature probability distribution* $P(T)$ produced by single-photon (stochastic) heating,

$$\varepsilon_\lambda = \int da \frac{dn}{da} \pi a^2 Q_{\text{abs}}(a, \lambda) \int dT P(a, T) \pi B_\lambda(T), \quad (9)$$

plus an equilibrium term for large grains. The emission in each cell is then normalized so that its integral equals the locally absorbed power, which makes energy conservation exact regardless of the SEDust internal normalization. The emitted spectrum is resampled onto the transport grid, and the emission is either integrated directly (the intrinsic SED) or transported as a second batch of packets to form the emergent SED and the pixel-resolved emission image.

Speed options. The exact stochastic solve is $\mathcal{O}(0.1\text{s})$ per cell. `par%dust_single_teq` replaces $P(T)$ by a single mixture equilibrium temperature (fast, no PAH/stochastic features). `par%dust_fast_table` builds a reference emission table over the field intensity $U = \int J_\lambda d\lambda / \int J_\lambda^{\text{ref}} d\lambda$ once, then interpolates each cell in U ($\sim 1\%$ SED-shape error, keeps PAH/stochastic).

Iteration. With `par%dust_niter` > 1 the emitted packets are transported so that dust self-absorption heats other cells; the field $J_\lambda \rightarrow J_\lambda^{\text{stellar}} + J_\lambda^{\text{dust}}$ and the emission is recomputed until the total emitted luminosity converges to `par%dust_tol`. This matters only at high infrared optical depth.

Grain models. `par%dust_model_sed` selects `astrodust` (Hensley & Draine 2023), `dl07` (Draine & Li 2007), or `zubko` (Zubko et al. 2004 BARE-GR-S).

5.2 Mode 2: Bjorkman & Wood (2001)

The immediate-reemission scheme uses fixed-energy packets and a mixture mean opacity. When a packet is absorbed in a cell, the cell's absorbed energy increases, raising its equilibrium temperature T ; the packet is re-emitted at once with a wavelength drawn from the *temperature-correction* spectrum

$$\frac{dP}{d\lambda} \propto \kappa_{\text{abs}}(\lambda) \frac{\partial B_\lambda(T)}{\partial T}, \quad (10)$$

which corrects the previously-emitted spectrum for the temperature increment so that, in steady state, the cell radiates a $\kappa_{\text{abs}} B_{\lambda}(T)$ spectrum. No iteration and no radiation-field storage are needed; the method is approximate (equilibrium, mixture opacity, no PAH/stochastic features) but cheap. The emergent dust light lands directly in the observer Scattered SED.

6. Modified Random Walk

In cells that are very optically thick to *scattering* a packet takes many tiny steps. The Modified Random Walk (Min et al. 2009; Robitaille 2010) replaces them with a single diffusion step. A sphere of radius R_0 (the distance to the nearest cell wall) is inscribed; the packet is moved to a random point on that sphere, and the physical path length travelled inside is sampled from the tabulated diffusion first-passage distribution $\zeta(y)$. MoCafe diffuses with the *scattering* opacity and decays the weight by the absorption along the path, $w \leftarrow w e^{-a_{\text{abs}} c t}$, and applies Russian roulette (survival probability RR_SURV below a cutoff RR_CUT) so that high-albedo, high- τ clouds terminate. MRW triggers when the nearest-wall scattering optical depth exceeds `par%mrw_gamma`; its speed-up is large only for near-pure-scattering, very thick cells.

7. Source (galaxy) models

A stellar component emits packets from a spatial distribution. Multiple components (`par%nsource`) are chosen in proportion to luminosity.

7.1 Exponential disk

A razor-thin exponential disk has surface density $\Sigma(r) \propto e^{-r/r_s}$, so the probability of a packet at cylindrical radius r is $p(r) \propto 2\pi r \Sigma(r) \propto r e^{-r/r_s}$. This is the Gamma(2) (double-exponential) law, sampled exactly by `rand_r1exp` (the sum of two exponential deviates). The vertical structure is exponential (`rand_zexp`) or sech^2 (`rand_sech2`).

7.2 Spiral arms

Log-spiral arms multiply the disk by

$$1 + A \sin \left[m \left(\frac{\ln r}{\tan p} - \phi \right) \right], \quad (11)$$

with m arms, pitch angle p , and amplitude A . For *sources* the azimuth is drawn by rejection against this factor (accept if $(1+A)\xi \leq 1 + A \sin[\dots]$); for the *dust* density the cell opacity is multiplied by the same factor. Seen face-on, the absorbed-power map traces the arms (validated in `examples/galaxy/spiral_faceon.in`).

7.3 Sersic bulge (deprojection + alias sampling)

The projected 2-D Sersic profile is $I(R) = I_0 \exp[-b_m (R/R_e)^{1/m}]$, with b_m from Ciotti & Bertin (1999). MoCafe emits from the 3-D luminosity distribution, obtained by deprojecting $I(R)$ with an Abel transform; the deprojected cumulative

luminosity within spherical radius r is

$$\mathcal{L}(< r) = \gamma(2m+1, b_m r^{1/m}) + \frac{2}{\pi} \frac{b_m^{2m+1}}{m \Gamma(2m+1)} r^2 \int_1^\infty x f(x) e^{-b_m (rx)^{1/m}} (rx)^{1/m} dx, \quad (12)$$

computed once on a log-radius grid (`sersic_cumulative_3D`). The radius is then drawn with the *alias* method (§8.) rather than an inverse-CDF search: the bin probabilities (differences of the cumulative) build an alias table, and `rand_alias_linear` returns a continuous radius by linear interpolation within the selected bin. The point is placed isotropically and flattened in z by the axial ratio. The alias-sampled distribution reproduces the analytic cumulative to $\sim 10^{-4}$.

7.4 Boxy / bar / x-bar bulges

Generalized-ellipsoid bulges use the radius $r = (|x|^n + |y|^n + |z|^n)^{1/n}$ with boxiness n ($n = 2$ ellipsoid, $n > 2$ boxy) and a power-law (boxy), Gaussian (bar), or peanut (xbar) profile. These are drawn with a slice-within-Gibbs sampler (`random_bulge.f90`): an auxiliary level is drawn under the current density and each coordinate is updated uniformly on the resulting slice. Consecutive draws are therefore correlated—adequate for a bulge, but statistically weaker than the independent alias-sampled Sersic.

8. Random sampling

8.1 Alias method

Walker’s alias method samples an index from a discrete distribution $\{p_k\}$ in $\mathcal{O}(1)$. A setup pass (`random_alias_setup`) converts $\{p_k\}$ into an acceptance probability q_k and an alias A_k per bin; a draw picks a random bin k and returns k with probability q_k , else A_k . MoCafe uses it for wavelength sampling, the Mueller S_{11} , and the Sersic radius.

8.2 Continuous alias (piecewise-linear)

For a continuous variable tabulated at nodes x_i with density p_i , the bin selection probability is the trapezoid area $\frac{1}{2}(p_i + p_{i+1})(x_{i+1} - x_i)$; after the alias picks a bin, the value is returned by inverting the linear density inside it (`rand_alias_linear`). This gives an independent, continuous draw and is used for external-illumination angles, the scattering angle, and the Sersic radius.

9. HEALPix all-sky interior observer

For an observer *inside* the medium (the Milky-Way case) there is no single image plane. Each scattering / dust-reemission / direct event peels toward the observer point, and the contribution is binned into the HEALPix pixel (RING scheme, $n_{\text{pix}} = 12 n_{\text{side}}^2$, equal-area pixels $\Omega = 4\pi/n_{\text{pix}}$) of the arrival direction $\hat{\mathbf{n}}$, computed by `vec2pix`. The optical depth to the observer is integrated along the matching ray (Cartesian or AMR). The output carries the standard HEALPix keywords so it reads directly with `healpy`. Because each event maps to the single pixel of its *source* direction, a diffuse map needs

many events distributed over the sky; concentrated (inner-disk) far-infrared emission is sparse at high n_{side} and benefits from a finer spatial grid or smoothing.

10. Energy conservation

Energy conservation is the main internal check. The total stellar luminosity equals the escaped starlight plus the absorbed (hence re-emitted) power; the dust solvers normalize the emission to the absorbed power cell by cell, so the integrated intrinsic dust luminosity equals L_{ABS} , and the emergent SED integrated over $4\pi d^2$ recovers the same value to Monte-Carlo accuracy. The `_jlam` output additionally reports the pathlength-tallied absorbed power against an independent event counter.